Recurrent Neural Networks versus HAR for Five-Day-Ahead Volatility Forecasts of Equities and Bitcoin

hypoq autonomous research harness[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: an LSTM and the HAR benchmark have equal expected QLIKE loss for the 5-day-ahead variance; H_1: the LSTM has lower loss
Data: Daily OHLC of the S&P 500 and Bitcoin; S&P 500 test 2023-01-03 to 2026-10-16 (951 days); Bitcoin test 2023-01-01 to 2026-10-20 (1,389 days)
Method: Garman–Klass variance, HAR on log variance, LSTM on 22-day windows with early stopping and 5 seeds, chronological split at 2023-01-01, one-sided DM test with HAC errors, Bonferroni $\alpha = 0.025$
Results: S&P 500: QLIKE HAR 0.2263 vs LSTM 0.2348 (DM $p = 0.742$); Bitcoin: QLIKE HAR 0.2727 vs LSTM 0.2361 (DM $p = 0.000$). Decision: **INCONCLUSIVE**

1 Design

The target is the log mean Garman–Klass variance of the next five sessions. HAR regresses it on the current, weekly and monthly log variance [Corsi, 2009]; the LSTM sees the same information as 22-day windows. Both receive the same log-normal bias correction when converted to variance. Training labels that overlap the test period are purged.

2 Results

3 Limitations

Garman–Klass from daily bars is a noisy proxy for realised variance; one chronological split is used rather than rolling re-estimation; the network is small and its hyper-parameters were not tuned on the test period.

Table 1: Out-of-sample forecast loss (lower is better) and one-sided Diebold–Mariano test, H_1: LSTM better

Asset	QLIKE HAR	QLIKE LSTM	MSE-log HAR	MSE-log LSTM	DM t	p
S&P 500	0.2263	0.2348	0.3836	0.3902	-0.65	0.7417
Bitcoin	0.2727	0.2361	0.4525	0.4146	3.76	0.0001

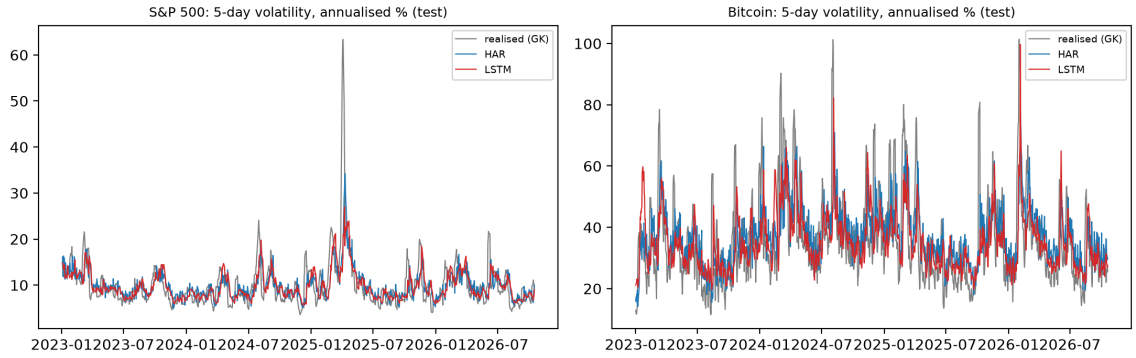


Figure 1: Test-period forecasts.

References

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.